

- 1) Total mass of MW and M31 compare: MW and M31 have the same total mass despite different mass distributions
Galaxy dominates the total mass: Dark matter halos $\rightarrow$ MW

$$M_W = 0.085$$

- 2) Stellar mass of MW and M31 compare: Disk to bulge $\rightarrow M_{31} = 0.139 \leftarrow$ M31 has more stellar mass than MW
Galaxy expected to be more luminous: M31 since it has more stellar mass so its potentially more bigger

$$\frac{M_W}{M_{31}} = \frac{1.975}{1.921} = 1.028$$

- 3) Total dark matter mass of MW and M31 compare (ratio): $\frac{M_W}{M_{31}} = \frac{1.975}{1.921} = 1.028 \leftarrow$ they are close in value

Surprising - their difference in stellar mass: Yes since M31 has more stellar mass and is more luminous: Formation/ history could have played a part.

- 4) Ratio of stellar mass to total mass for each galaxy (Baryon fraction): $M_W = 0.036, M_{31} = 0.058, M_{33} = 0.046$

$\frac{\rho_b}{\rho_m} \sim 16\%$ $\leftarrow$ locked in baryons. Ratio compared to the baryon fraction for each galaxy: All ratios are lower than the universe

with M31 being the closest. The universal baryon force differs from the galaxies: Formation/evolution of about

anything varies as they might take different steps/processes than expected. In this case it leads to different

baryon fractions than the universe as a whole. M31 seemed to not lose gas compared to the other galaxies.